



Ministry of
IT & Telecom



Hands-on Exercise No. 02
DigiSkills 3.0 Batch-04
Artificial Intelligence using Python

Total Marks: 10

Due Date: 10/09/2026

Instructions:

Please read the following instructions carefully before submitting this Hands-on Exercise:

- Use MS Word to prepare exercise solution.
- You may consult tutorials and videos if the concept is not clear.
- Your submitted exercise will not be considered/counted if:
 - It is submitted after due date.
 - It is not in the required format (doc or .docx)
 - It does not open, or file is corrupt.
 - It is copied (partial or full) from any source (websites, forums, students, etc.)

Learning Outcome:

After completing this exercise, you shall be able to:

- **Load and explore a real-world dataset** using Pandas by examining its structure, variables, statistics, and data distributions.
- **Clean and prepare data for machine learning** by handling missing values, identifying relevant features, and preparing the target and independent variables.
- **Build, evaluate, and analyze an SVC classification model** using appropriate performance metrics and visualizations to understand passenger survival patterns.

Problem Statement

In this hands-on exercise, you are required to work with the **Titanic dataset** to perform data exploration, cleaning, feature preparation and visualization using Python. You will load the dataset into a DataFrame, explore its numerical and categorical variables, handle missing values, identify the target and independent features, and build an **SVC classification model** to predict passenger survival. You will then evaluate the model using accuracy, confusion matrix, precision, recall, and F1-score, and create visualizations to analyze survival patterns based on gender, passenger class, and fare. The objective is to understand how a real-world dataset can be explored, prepared, analyzed, and used to build and evaluate a machine learning classification model.

You can use **Jupyter Notebook, Google Colab, VS Code, or any other Python programming environment** to write and execute your code. After completing each task, take screenshots of your **code and its output** and paste them into a single **MS Word (.doc/.docx) file** for submission.

Dataset Description:

The **Titanic dataset** contains information about passengers who travelled on the RMS Titanic, which sank on April 15, 1912, after colliding with an iceberg. The dataset includes details such as **passenger class, gender, age, number of siblings/spouses, number of parents/children, ticket fare, and port of embarkation**. It also contains the **Survived** variable, which indicates whether a passenger survived the disaster (1) or did not survive (0). In this exercise, the dataset will be used to explore the factors related to passenger survival and to build a **Support Vector Classifier (SVC)** that predicts whether a passenger survived.

Download the **titanic.csv** file from LMS using the following link:

<https://lms.digiskills.pk/Courses/AIP301/Downloads/titanic.csv>

Tasks:

1. Dataset Loading and Initial View:

(Marks 2)

- Import the necessary Python libraries and load the dataset into a Data Frame.
- Display the total number of rows and columns in the concern dataset.
- Display the first 7 rows of the dataset.

2. Data Exploration:

(Marks 2)

- Calculate and display the descriptive statistics (count, mean, standard deviation, minimum, maximum, and quartiles) for the numerical variables.
- Identify and display the numerical and categorical variables present in the Titanic dataset.
- Analyze categorical variables such as Sex, Passenger class by displaying their unique values and frequency distributions.

3. Data Cleaning:

(Marks 2)

- Count the missing values present in each column.
- Fill the missing values using the mean or median.
- Verify that all missing values have been handled successfully.

4. Feature Selection & Renaming:

(Marks 2)

- Identify the target (dependent) variable and the independent features
- Divide the dataset into training and testing sets using an appropriate train-test ratio.
- Build and train an SVC model using the training data. Make predictions on the test data and evaluate the model using accuracy, confusion matrix, precision, recall, and F1-score.

5. Data Visualization:

(Marks 2)

- Create an appropriate visualization to analyze the combined relationship between Gender, Passenger Class, and Survival.
- Visualize the distribution of passenger **Fare** using a histogram or box plot. Identify any noticeable skewness or extreme values.
- Create an appropriate visualization to compare **survival and non-survival among male and female passengers**. Analyze which gender had a higher survival rate.

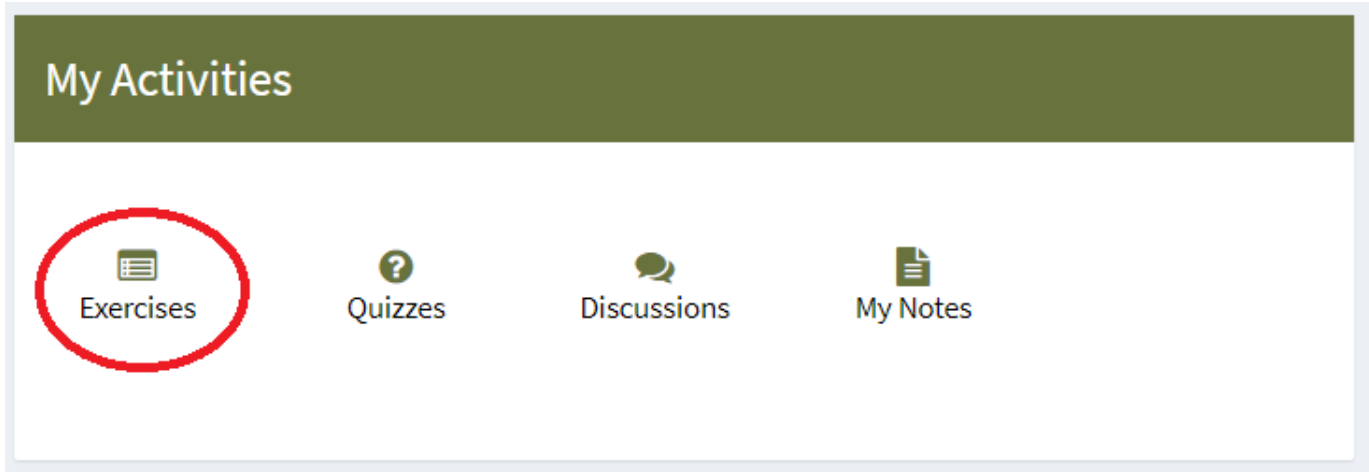
NOTE: In this solution file, no need to send/attach code file; only provide screenshots of the above tasks along with code outputs.

BEST OF LUCK ☺

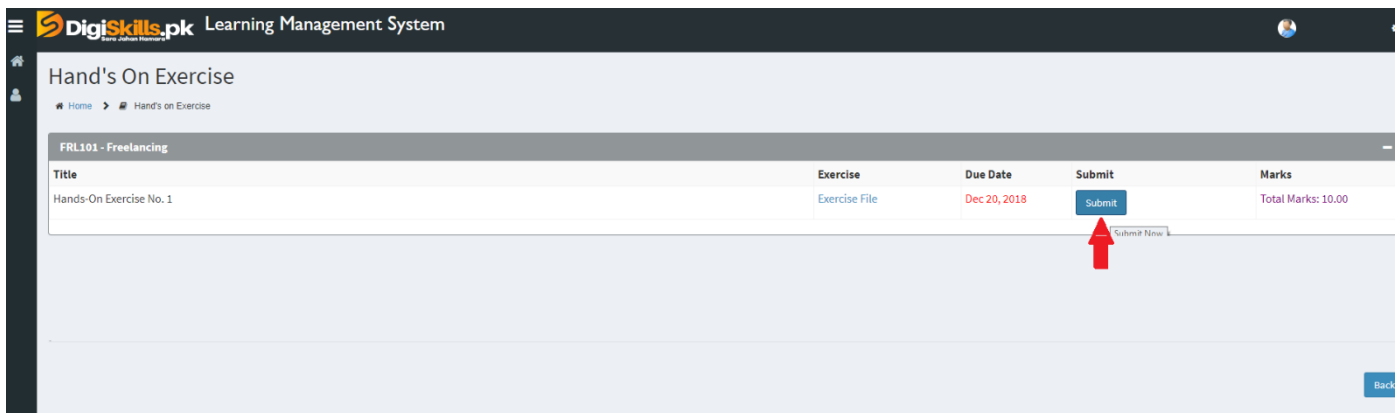
How to submit solution file on LMS?

Please perform the following steps for submitting your solution using LMS:

- 1) Login to the LMS
- 2) Click on the **Exercises** button within the **My Activities** section



- 3) Click on the submit button to upload your Solution.



- 4) Keep in mind to upload your Solution in .doc or .docx format